THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): A body of mass 5 kg has the same mass on the Moon. Reason (R): Mass depends on the gravitational pull at that place. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): A drinking straw can lift liquid into the mouth. Reason (R): Atmospheric pressure pushes the liquid up when pressure in the straw is reduced. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): The temperature stays constant while ice melts. Reason (R): The heat supplied is used to increase the kinetic energy of particles. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): Sodium is stored under kerosene. Reason (R): Sodium is a very reactive metal. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Assertion (A): Powdered sugar dissolves faster than sugar crystals. Reason (R): Powder has a smaller surface area. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A 60 kg astronaut on the Moon has mass: 1

(a) 360 kg **(b)** 10 kg **(c)** 0 kg **(d)** 60 kg

7. A block of weight 60 N rests on a table. The force exerted by the table on it is: 1

(a) 60 N upward **(b)** 30 N **(c)** 0 **(d)** 60 N downward

8. Ice at 0 degrees Celsius and water at 0 degrees Celsius are compared. Which statement is correct? 1

(a) Particles have the same kinetic energy but different arrangement

(b) Water has fewer particles

(c) Ice particles move faster

(d) Water has lower temperature

9. The total number of atoms in 2 molecules of H_2SO_4 is: 1

(a) 7 **(b)** 28 **(c)** 4 **(d)** 14

10. From 500 g of 20 per cent sugar solution, 100 g of water is evaporated. The new percentage is: 1

(a) 10 per cent **(b)** 30 per cent **(c)** 25 per cent **(d)** 20 per cent

11. To reduce the wear of the surfaces of the moving parts of a machine, we should: 1

(a) use a rough material **(b)** apply a lubricant

(c) increase the load **(d)** stop the machine

12. A coastal village plants mangroves to: 1

(a) stop wind completely **(b)** increase rain

(c) create wind **(d)** reduce the force of the surge

13. **Assertion (A):** The walls of a dam are made thicker at the bottom. 1

Reason (R): Pressure of water increases with depth.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** A drop of ink spreads faster in hot water than in cold water. 1

Reason (R): The particles of matter are continuously moving and move faster at a higher temperature.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** Metals are generally good conductors of heat and electricity. 1

Reason (R): Metals are malleable and ductile.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. State why heavy trucks have more tyres. 2

17. Why is there a risk of a roof being blown off in a strong storm? 2

18. State why evaporation is called a surface phenomenon. 2

19. Why are noble gases unreactive and how is this useful? 2

20. Why does tea made with the same sugar but less water taste sweeter? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Why is sea water saltier than river water? 3

22. Explain why a cyclist bends forward when riding against the wind. 3

23. Compare the pressures exerted by 3
(a) a 70 kg person (about 700 N) on both feet (0.04 square metre) and
(b) the same person on skis of total area 0.4 square metre. Why is it easy to walk on snow with skis?

24. Explain the statement: 'Temperature is a measure of the average kinetic energy of particles' using hot and cold water. 3

OR

Compare the properties of sodium, chlorine and sodium chloride to show that a compound is different from its elements. 3

25. A sample of 200 g of a 30 per cent sugar solution is diluted to 12 per cent. Find the water that must be added. 3

OR

How can a magnetised needle be made? Describe the process. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Isha adds one drop of blue ink to each of three equal beakers of water at 5, 25 and 65 degrees Celsius without stirring. After one minute the colour has spread through the hottest beaker, partly through the middle beaker and only a little in the coldest beaker. 5
(i) Name the process of spreading of ink on its own.
(ii) In which beaker was it fastest?
(iii) Which property of particles explains the spreading?
(iv) Why is the spreading faster at higher temperature?

27. A man of weight 700 N stands in a lift at rest. Discuss the forces. Compare the forces when the lift moves up with constant speed and explain why the reading of a weighing machine is unchanged. 5

OR

A man weighs 800 N and a cow's hoof area is 0.01 square metre for a 4000 N cow with four hooves. Compare the pressure beneath the cow's hooves with the man's pressure when standing on two feet of total area 0.04 square metre. 5

END OF QUESTION PAPER — Total: 50 Marks